

**1. Show that there is an in-order enumerator (that will enumerate all strings in L in lexicographic order) for a language L if and only if that language is decidable.**

If L is decidable by some TM M, the enumerator operates by generating the strings in lexicographic order, testing each in turn for membership in L using M, and printing the string if it is in L. If L is enumerable by some enumerator E in lexicographic order, we consider the following two cases:

- If L is finite, it is decidable because all finite languages are decidable, just hardwire each of the strings into the TM M.

E = "On any input,

Ignore the input

1. For  $i = 1, 2, \dots n$ .

2. Run M on the  $i^{th}$  string in Sigma-star, according to the desired ordering

3. If M accepts, print the string"

- If L is infinite, a TM M that decides L operates as follows:

On receiving input w:

M = "On input w,

1. Run E

2. For every string s printed by E, if  $s = w$ , accept w

3. Else if  $s < w$  in the desired ordering, continue

4. Else if  $s > w$ , reject w"

A finite number of strings of L are  $\leq w$  in the desired ordering. After E prints a finite number of strings, we can decide if w is in L or not. Therefore, this is guaranteed to halt, since E never runs out of strings and is finite.

**1. Let  $S = \{ \langle M \rangle \mid M \text{ is a DFA that accepts } w^R \text{ whenever it accepts a string } w \}$ .**

For any language L, let  $L^R = \{ w^R \mid w \text{ is in } L \}$ .

If  $\langle M \rangle$  is in S, then  $L(M) = L(M)^R$ .

The following TM T decides S.

T = "On input  $\langle M \rangle$ , where M is a DFA:

1. Construct DFA N that recognizes  $L(M)^R$  using the following construction.

- A NFA N' is constructed from M first.

- Keep all states in M and reverse the direction of all arrows in M.

- Set the start state in M as the accept state in N'.

• Create a new state as the start state in N' and introduce new  $\epsilon$ -transition arrows from the start state to every accept state in M.

- Transform the NFA N' to a DFA called N, which recognizes  $L(M)^R$

2. Run TM D, obtained from slides, on  $\langle M, N \rangle$ , where D is the TM deciding  $EQ_{DFA}$ ."

3. If D accepts, accept, else reject.

The TM  $EQ_{DFA}$  is decidable, and therefore, D will always halt. This makes T always halts, and therefore, S is decidable.

**2. Consider the problem of deciding whether the language given by a CFG over  $\{0,1\}$  contains a string not of the form  $1^*$  (i.e. does it contain a string that contains at least one symbol '0'?).**

Assume that  $G$  is over  $\{0,1\}^*$ . Then, we need to show that the language  $A = \{\langle G \rangle \mid G \text{ is a CFG over } \{0,1\}^* \text{ and } 1^* \cap L(G) \neq \emptyset\}$  is decidable. Since  $1^*$  is regular and  $L(G)$  is CFL then  $1^* \cap L(G)$  is a CFL. Hence, the TM  $M$  that decides  $A$  is:

$M =$  “On input  $\langle G \rangle$  where  $G$  is a CFG:

- Construct CFG  $H$  such that  $L(H) = 1^* \cap L(G)$
- Run the TM  $R$  that decides the language  $E_{CFG}$  on  $\langle H \rangle$
- If  $R$  accepts, reject; if  $R$  rejects, accept.

**3. Let  $L_{11} = \{\langle M \rangle \mid M \text{ is a TM that accepts the string } 11\}$ .**

**Show that  $L_{11}$  is un-decidable by giving a many-reduction from the halting problem.**

1)  $ATM = \{\langle M, w \rangle \mid M \text{ is a TM and } M \text{ halts on input } w\}$  is not decidable.

The TM  $U$  recognizes  $ATM$ .

$U =$  “On input  $\langle M, w \rangle$  where  $M$  is a TM and  $w$  is a string:

Simulate  $M$  on  $w$

- If  $M$  ever enters its accepts state, accept.
- If  $M$  ever enters its rejects state, reject.
- If  $M$  loops on  $w$ , then  $U$  loops on  $\langle M, w \rangle$ , which is why is not a decider.
- $U$  cannot detect that  $M$  halts on  $w$ .

$ATM$  is also known as the halting problem.

$U$  is known as the Universal TM because it can simulate every TM, including itself.

2) To prove that  $L_{11} = \{\langle M, w \rangle \mid M \text{ is a TM and } M \text{ halts on input } 11\}$  is not decidable. I will use the idea that “If  $A$  is not decidable and reducible to  $B$ , then  $B$  is not decidable.”

Suppose  $R$  decides  $L_{11}$ . We construct  $S$  to decide  $ATM$ .

$S =$  “On input  $\langle M, w \rangle$

Run  $R$  on input  $\langle M, w \rangle$

- If  $R$  rejects reject.
- If  $R$  accepts, simulate  $M$  on  $w$  until it halts,  $w = 11$ .
- If  $M$  has accepted, accept. If  $M$  has rejected, rejects”.

Since  $ATM$  is reduced to  $L_{11}$ ,  $L_{11}$  is not decidable.